



**MINISTRY OF STATISTICS & PROGRAMME IMPLEMENTATION
NATIONAL SAMPLE SURVEY OFFICE
DATA PROCESSING DIVISION
TECHNICAL COORDINATION**

PROJECT REPORT FOR THE SUMMER INTERNSHIP SCHEME 2024-25 OF MOS&PI

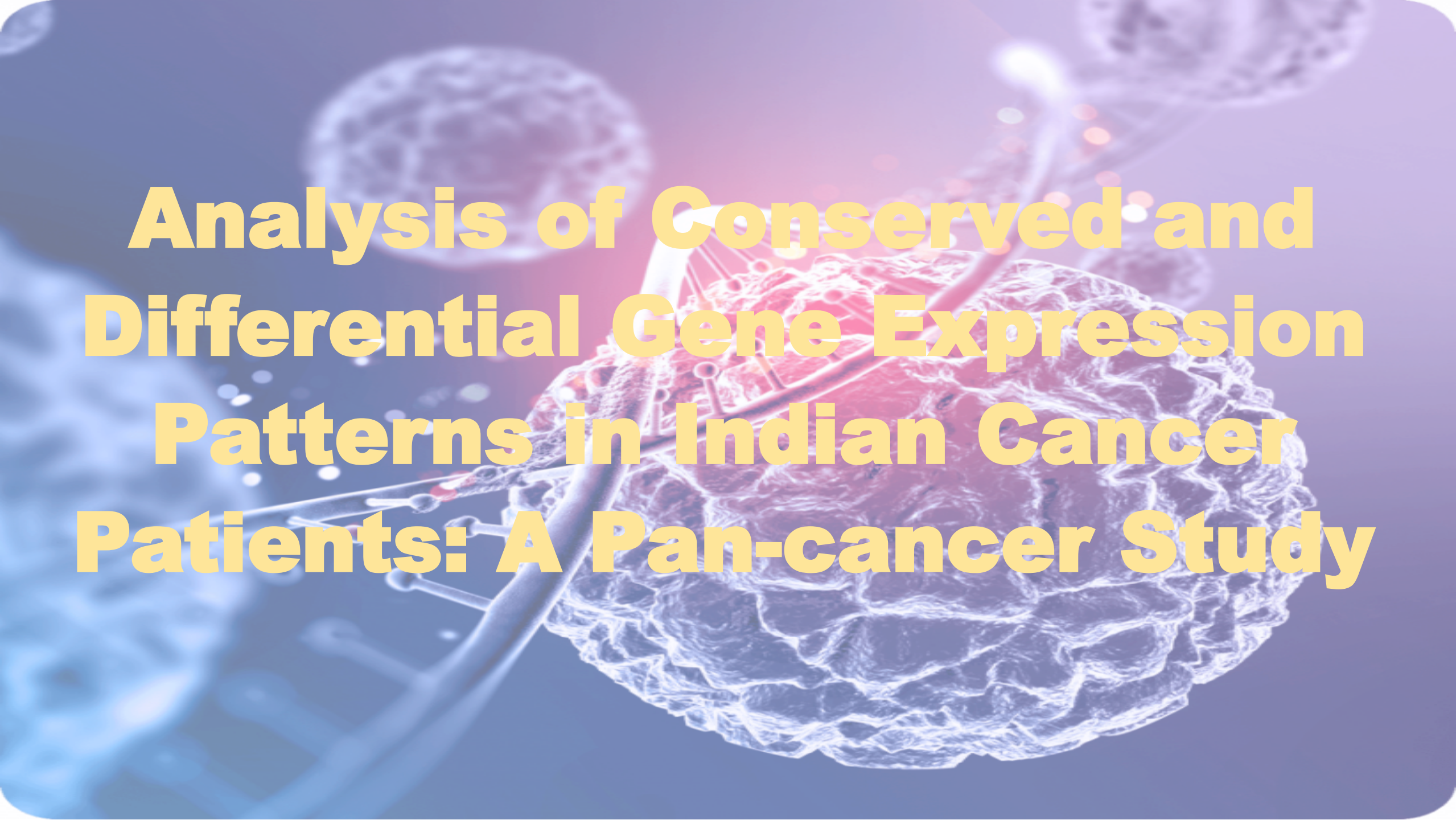
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SESSION:- 2022-2024

REGISTRATION NO. :- VB-0311 OF 2022-2023



Analysis of Conserved and Differential Gene Expression Patterns in Indian Cancer Patients: A Pan-cancer Study

Introduction

- ❑ Cancer is a multifaceted disease characterized by uncontrolled cell growth in any part of the body. The genetic mutations and variations that drive cancer development and progression **vary widely among individuals and populations.**
- ❑ Pan-cancer studies aim to uncover commonalities and differences across various cancer types. Specifically, they may reveal **conserved** gene expression patterns fundamental to cancer biology and identify **differential** gene expression patterns that may be unique to specific cancer types or populations.
- ❑ Indian Council of Medical Research (ICMR) aims to investigate various types of cancers, including breast cancer, renal cancer, colon cancer, lung cancer, and prostate cancer, which have become a growing concern in recent years. This research will facilitate early detection of each cancer type, thereby reducing the fatality rate.
- ❑ This study represents a significant step towards understanding the **genetic underpinnings of cancer in the Indian population** and has the potential to impact cancer treatment and research on a global scale.

Material and Methods

I. Data:

The dataset consists of 801 samples from 801 individuals diagnosed with various types of cancer. Each sample contains expression values for over 20,000 genes. The samples are categorized into one of the following tumours types:

BRCA (breast adenocarcinoma), COAD (colon adenocarcinoma), KIRC (kidney renal clear cell carcinoma), PRAD (prostate adenocarcinoma) and Lung adenocarcinoma (**LUAD**).

Source: collected the data from <https://www.kaggle.com/datasets/shibumohapatra/icmr-data/code>.

This data was update 2 years ago.

II. Methods

Mean-variance summary, low-varying covariate filtering, missing data cleaning, visual summaries of differential expression (scatter plot , boxplot etc), univariate multi-class logistic regression, principal component analysis, Cluster analysis etc.

❖ I used R programming for analysis the data of my project.



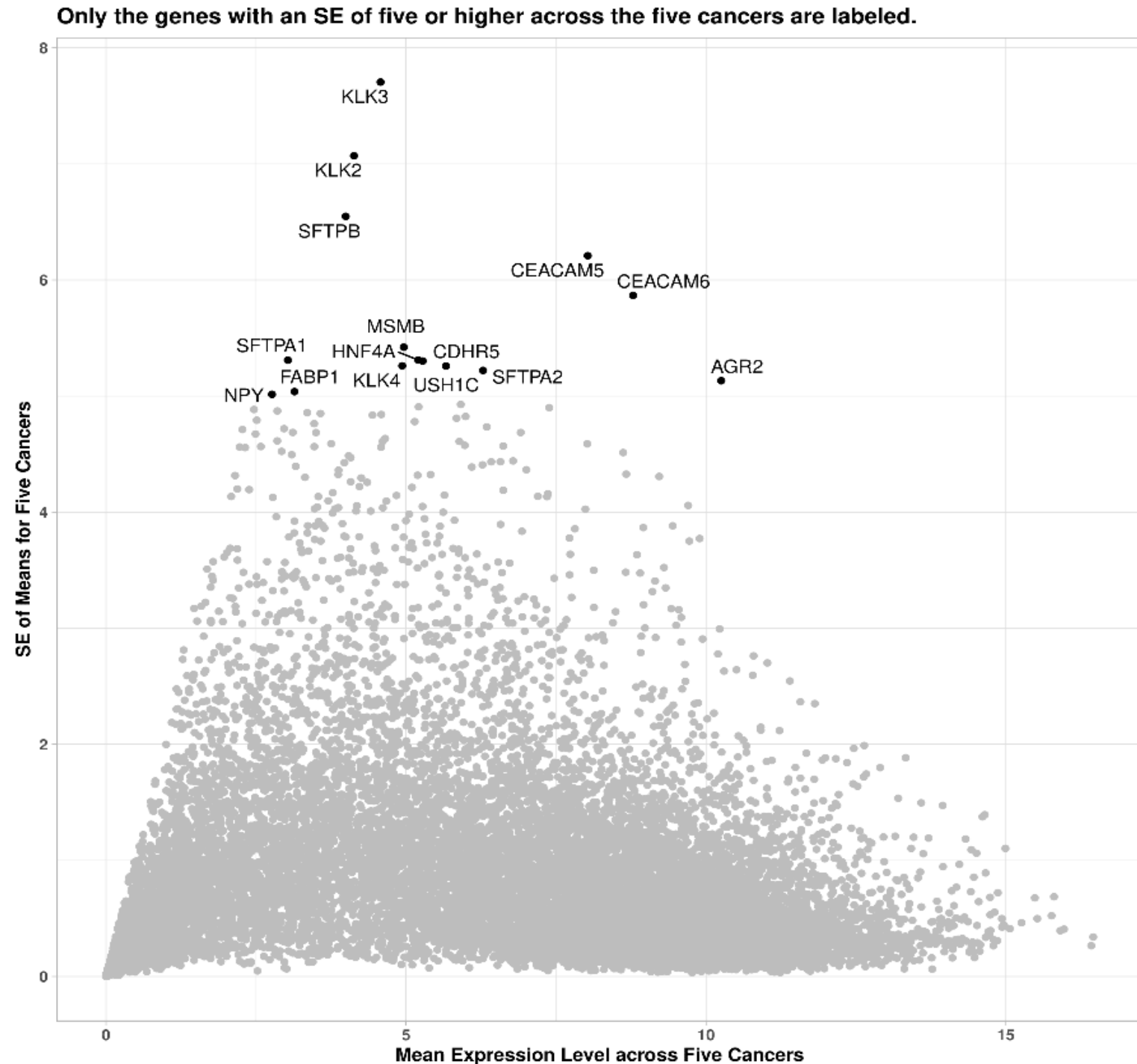
This data shows that, a particular expression of a particular gene of a particular cancer patient. That means for a cancer disease PRAD (prostate adenocarcinoma) of a patient sample_0 and for the gene A1BG the expression is 9.5135381.

	id	cancer	gene	expression
1	sample_0	PRAD	A1BG	9.5135381
2	sample_0	PRAD	A1CF	0.0000000
3	sample_0	PRAD	A2BP1	4.0636582
4	sample_0	PRAD	A2LD1	7.7648053
5	sample_0	PRAD	A2ML1	4.7476559
6	sample_0	PRAD	A2M	13.7143958
7	sample_0	PRAD	A4GALT	10.0344964
8	sample_0	PRAD	A4GNT	0.0000000
9	sample_0	PRAD	AAA1	0.0000000
10	sample_0	PRAD	AAAS	9.8334578
11	sample_0	PRAD	AACSL	0.0000000
12	sample_0	PRAD	AACS	10.8612653
13	sample_0	PRAD	AADACL2	0.0000000
14	sample_0	PRAD	AADACL3	0.0000000
15	sample_0	PRAD	AADACL4	0.5918709
16	sample_0	PRAD	AADAC	1.0102786
17	sample_0	PRAD	AADAT	6.9628497
18	sample_0	PRAD	AAGAB	10.9597050
19	sample_0	PRAD	AAK1	9.7695089
20	sample_0	PRAD	AAMP	12.2454042
21	sample_0	PRAD	AANAT	0.5918709
22	sample_0	PRAD	AARS2	8.9688280
23	sample_0	PRAD	AARSD1	10.1753000
24	sample_0	PRAD	AARS	12.3990643
25	sample_0	PRAD	AASDHPPT	8.4798772
26	sample_0	PRAD	AASDH	7.3557651
27	sample_0	PRAD	AASS	8.3310513
28	sample_0	PRAD	AATF	10.6434594
29	sample_0	PRAD	AATK	6.8212255
30	sample_0	PRAD	ABAT	10.6374943

❖ From the plot, the gene which is high variability and low mean expression or, low variability and high mean expression or approximately same variability and mean expression that is not considering.

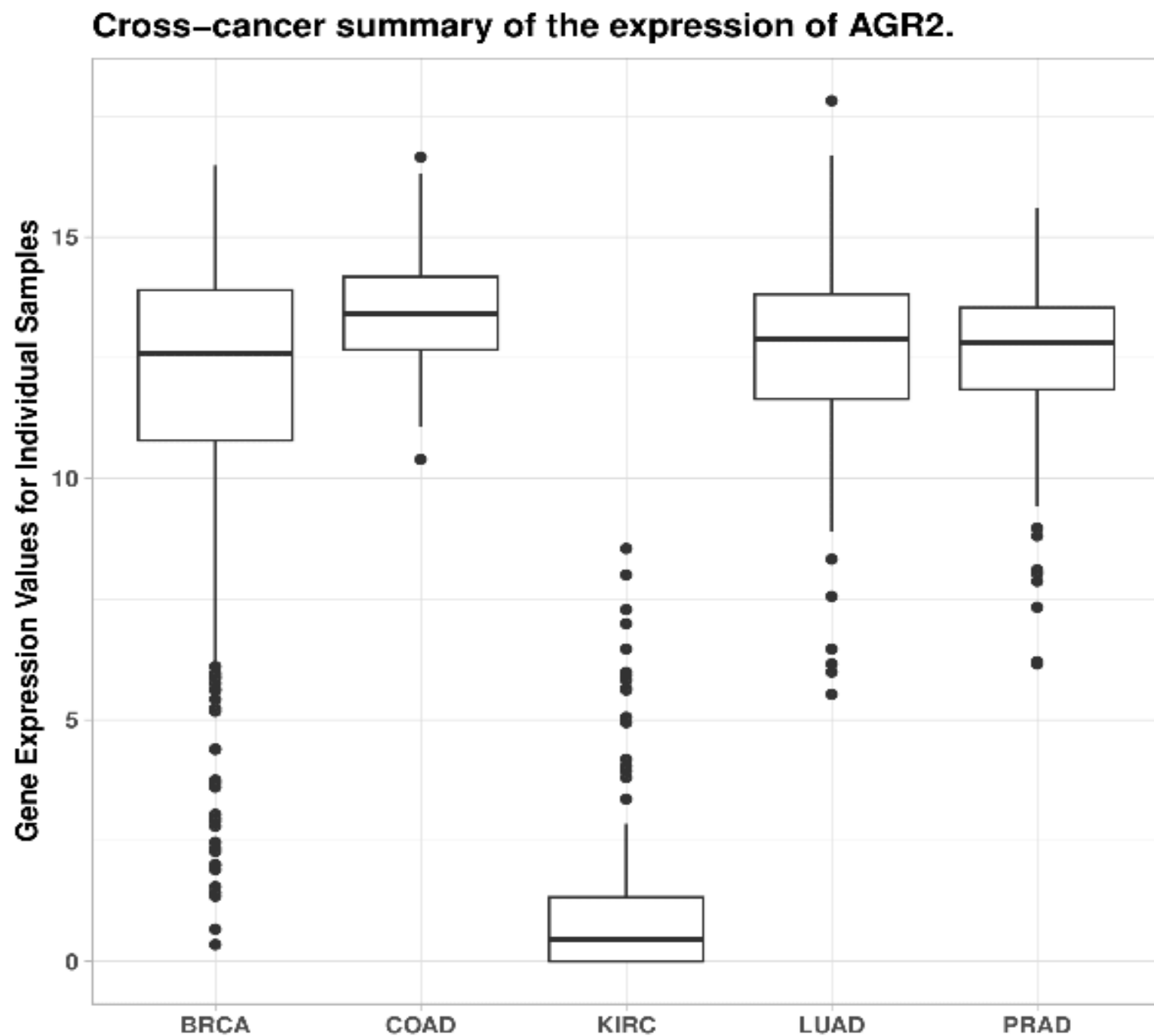
❖ I studied those gene of which Mean expression level across five cancer are grater than 5 and standard error of means for five cancers grater than 2.5 .

❖ Here we discussed that gene, which is highly expressed end high variability. The very important gene are KLK3, KLK2, SFTPB, CEACAM5, CEACAM6, MSMB, AGR2, CDHR5, NPY, KLK4, USH1C etc.

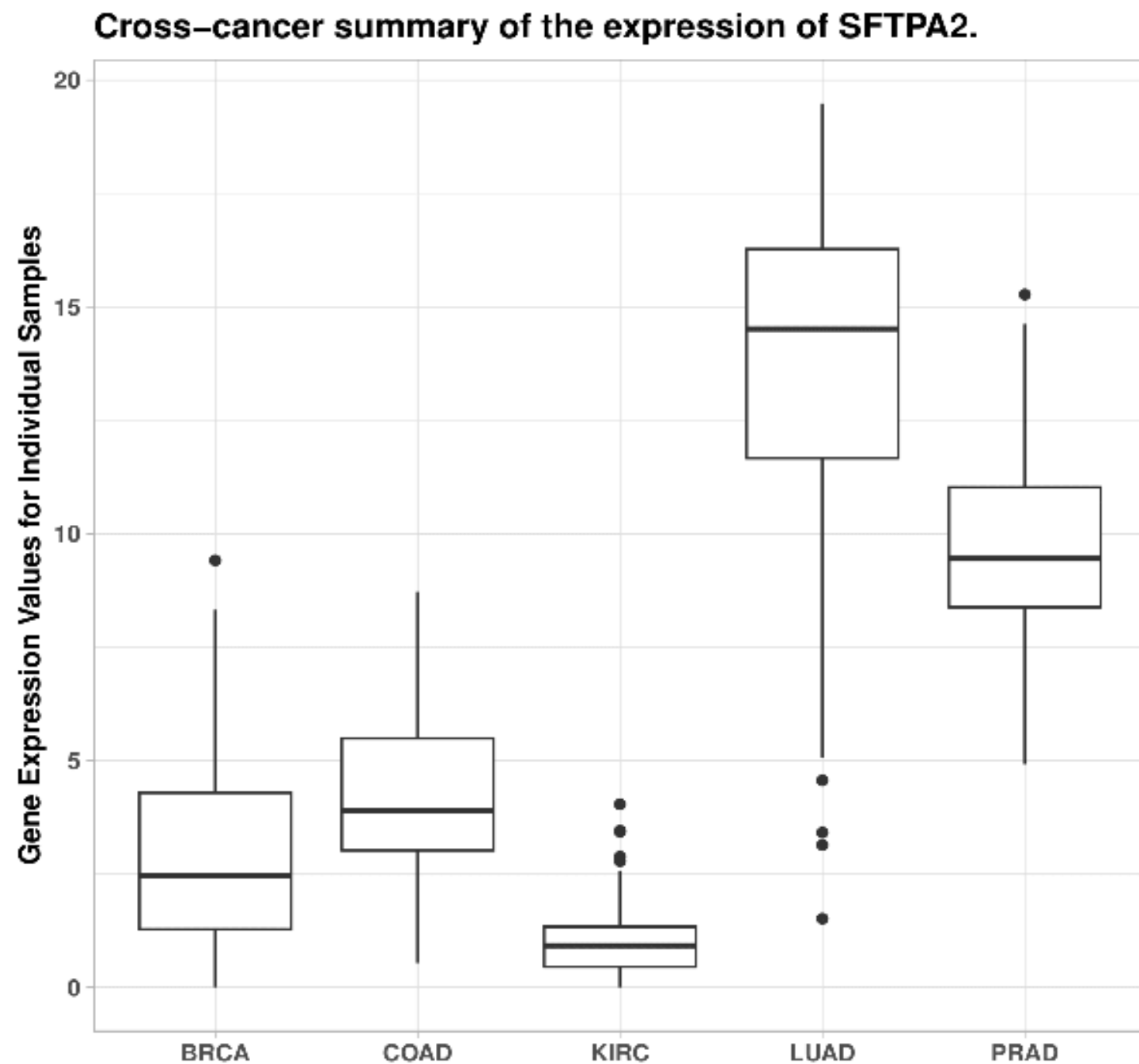


❖ This gene encodes a member of the protein disulfide isomerase (PDI) family, which are endoplasmic reticulum (ER) proteins involved in catalysing protein folding and thiol-disulfide interchange reactions.

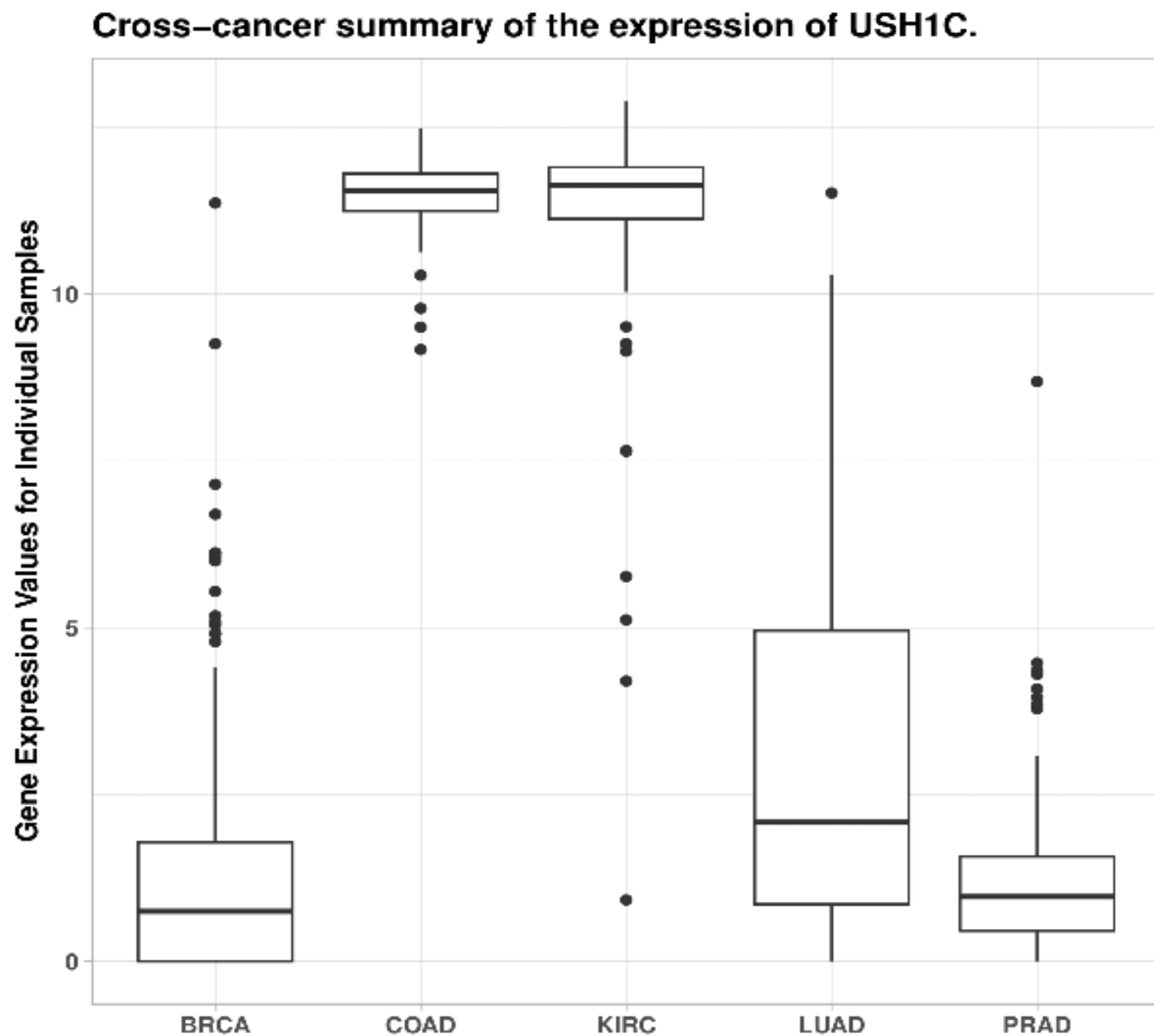
❖ Here we see high expression of this gene in all the adenocarcinomas but substantially low expression in the renal cell carcinoma.



- ❖ This gene is one of several genes encoding pulmonary-surfactant associated proteins (SFTPA) located on chromosome 10.
- ❖ Mutations in this gene and a highly similar gene located nearby, which affect the highly conserved carbohydrate recognition domain, are associated with idiopathic pulmonary fibrosis.
- ❖ This is in line with the figure beside– the gene shows high expression in the pulmonary cancer LUAD, but much lower expression in all the other cancers.



- ❖ USH1C (USH1 Protein Network Component Harmonin) is a Protein Coding gene for the protein Harmonin.
- ❖ The Human Proteome Atlas shows that in the TCGA dataset, this protein was found to be highly expressed in colon and renal cancers.
- ❖ Our gene-level findings are exactly the same, as can be seen in the beside figure.

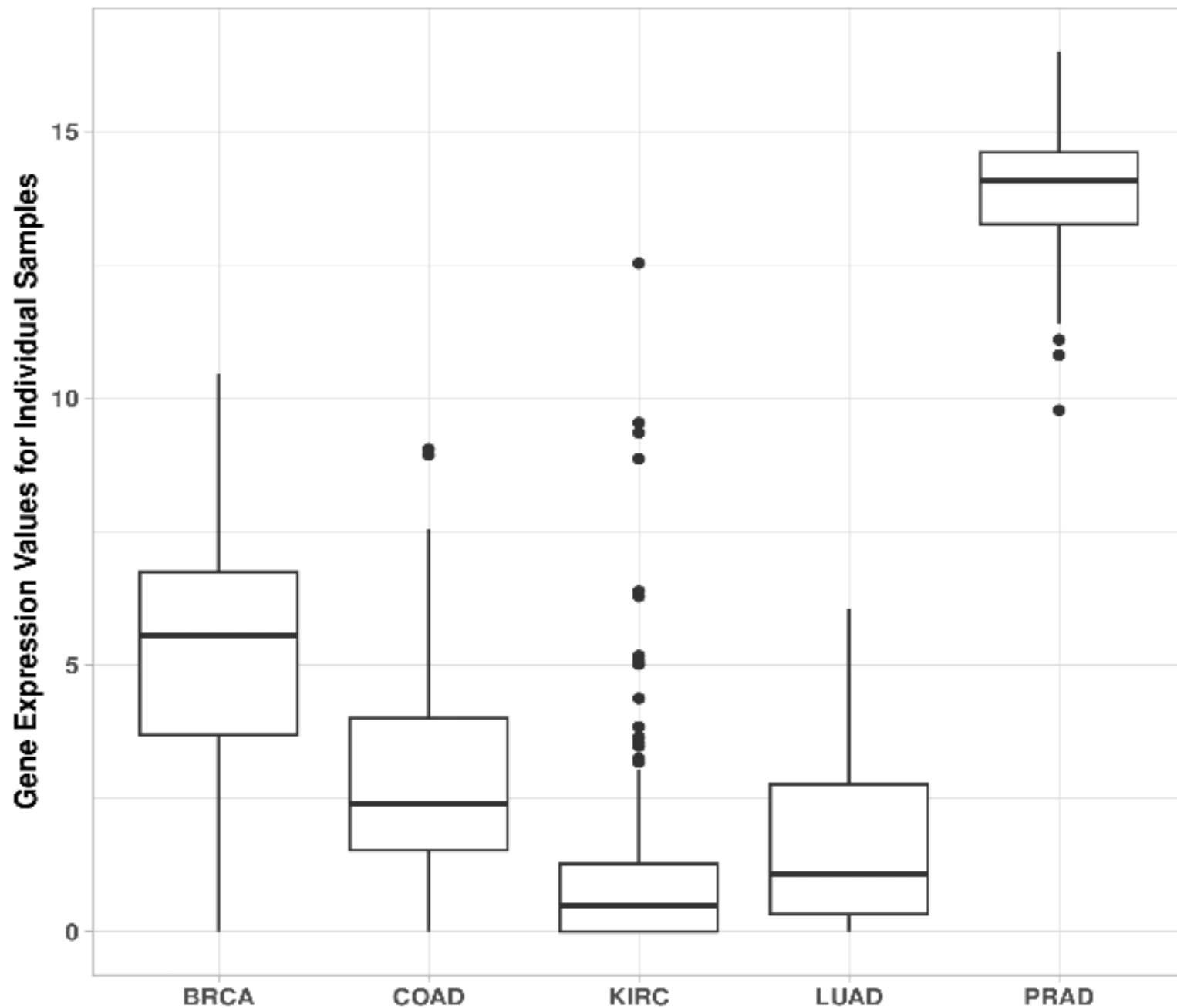


❖ Kallikrein-related peptidase 4 (KLK4), also known as prostase/KLK-L1, is a serine protease gene implicated in the development and progression of certain cancers.

❖ KLK4 is highly expressed in the prostate epithelial cells of both premalignant and malignant lesions, suggesting a role in prostate cancer biology.

❖ In our cohort, it is clearly showing highly outlying expression in the prostate samples compared to any other cancer.

Cross-cancer summary of the expression of KLK4.



- ❖ From the above scatter plot we can see that, five cancer represented by the five colour respectively.
- ❖ X axis represents the principal component 1 which variance 20.56% explained.
- ❖ Y axis represents the principal component 2 which is variance 12.9% explained.
- ❖ Every data points is a sample.
- ❖ Also, we can see that the two cancer KIRC (kidney renal clear cell carcinoma), PRAD (prostate adenocarcinoma) are separate from another cancer. The cancer BRCA (breast adenocarcinoma), COAD (colon adenocarcinoma), and Lung adenocarcinoma (**LUAD**) are similar type biologically.
- ❖ The principal component 1 separate the cancer KIRC from other four cancer.
- ❖ And the principal component 2 separate the cancer PRAD from other four cancer.



❖ KIRC is distinguished from the rest of the cancers by a cluster of 20 genes which are consistently strongly expressed in this cancer but not as strong in the others.

❖ On the other hand, the genes AGR2, FOXA1, and SPDEF are exhibiting low to no expression in the KIRC samples but relatively higher expression levels in the other cancers – specifically PRAD. These 23 genes have a profound effect on the first principal component, which results in the separation of KIRC from the other samples in the PCA scatterplot.



❖ PRAD is differentiated from the rest of the cancers by a cluster of 9 genes that are strongly expressed there but have average to low expression in the other cancers.

❖ These genes notably include the KLK class of genes which, as mentioned before, are known to be functional agents of relevance in prostate cancer.



Conclusion

- ❖ This study underscores such variability across cancers by assessing gene expression levels in Indian cancer patients from five distinct cancer types.
- ❖ We clearly observe differential patterns in specific genes and cluster of genes which make certain tissue-specific cancers different from the others (such as adenocarcinomas being different from other types of carcinomas).
- ❖ Using statistical techniques like PCA and clustering, we perform dimension reduction in high-dimensional expression dataset to identify key genes and gene groups of interest. We use modern visualization techniques to illustrate such differences.
- ❖ As a future direction, it would be interesting to explore the same comparisons in an integrative setting – combining data from other sources such as copy number alterations and protein expressions alongside gene expressions.

THANK YOU